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Prediction of a CLDN18.2 Targeted Antibody Drug Conjugate Pharmacokinetics in Cancer Patients Using PBPK Modeling and Simulation

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ABSTRACT

Antibody-drug conjugates (ADCs) represent a promising anticancer approach. Although physiologically based pharmacokinetics (PBPK) modeling became essential in Pharmacometrics to characterize exposure in different tissues, very few PBPK models have been published for ADCs, none within the PK-Sim/MoBi software. To capture the pharmacokinetics (PK) of an anti-Claudin 18.2 ADC, a PBPK model was built in PK-Sim and MoBi and compared to observations from three clinical studies after intravenous (IV) administration in 109 patients with cancer. The PK parameters were considered inaccurate if the predicted error ratios were outside the two-fold error range (0.5–2). In PK-Sim, we defined one PBPK model comprising three compounds (ADC, payload, and naked antibody), which were mechanistically linked. This model captured the ADC PK profile. However, additional clearance mechanisms were essential to improve the fit of the ADC elimination phase. After integration of target-mediated drug disposition (TMDD) and deconjugation of the payload in MoBi, 3 parameters were optimized for each of the ADC and the payload (degradation rate constant and reference concentration of the target, deconjugation rate constant, lipophilicity, non-specific hepatic clearance rate constant and passive renal clearance of the payload). The PK data were adequately captured for both observed compounds, with a predicted error ratio within the two-fold range: $C_{\max-ADC}$ (1.07–1.50), $C_{\max-Payload}$ (0.56–1.18), AUC_{0-504h_ADC} (0.73–1.23) and $AUC_{0-504h_payload}$ (0.77–1.37). The “parameter optimization” of different parameters allowed accurately capturing the observed data for both ADC and payload in cancer patients for an anti-Claudin 18.2 ADC. This analysis paves the way for PBPK modeling of other ADCs currently in development.

1 | Introduction

Antibody-drug conjugates (ADCs) became one of the fastest-developing classes in oncology in recent years because of their relatively low side effects compared to chemotherapies, broad treatment applications, and high therapeutic index [1]. ADCs

are biopharmaceutical products where a monoclonal antibody (mAb) is linked to a biologically active small molecule. As anticancer agents, the mechanism of action of ADCs is to release the cytotoxic drug (payload) inside the tumor cells to kill them without harming the healthy cells [1–3]. Upon binding to a target designed to be as specific as possible to the tumor cells

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Summary

- What is the current knowledge on the topic?
 - To our knowledge, no published model of ADC pharmacokinetics was developed and qualified with clinical data in the Open Systems Pharmacology Suite.
- What question did this study address?
 - Can we predict the plasma pharmacokinetics of targeted Claudin 18.2 ADC and the payload using physiologically based pharmacokinetic modeling and binding affinities for the target using an Open Source PBPK Platform such as PK-Sim?
- What does this study add to our knowledge?
 - Insights shared about the impact of model parameterization (such as TMDD and deconjugation processes) on PBPK predictive performance of both ADC and payload.
- How might this change drug discovery, development, and/or therapeutics?
 - It paves the way to the building and qualification of more complex PBPK models for drug modalities other than classical small molecules and mAbs.

via the fragment antigen-binding region (Fab) of the antibody, ADCs are internalized by tumor cells, in which they undergo deconjugation and lysosomal degradation, thus releasing the cytotoxic drug [4]. ADCs thus combine high specificity of mAbs with high potency of payloads in tumor tissues.

One of the challenges with ADCs is that the conjugation of the cytotoxic drug to the mAb often results in a heterogeneous mixture with a variable number of cytotoxic drugs attached to the mAb (drug-to-antibody ratio, DAR), and with different attachment locations on the antibody [5]. Another challenge is the deconjugation of the payload in plasma, which can result in important toxicity [1, 6]. To optimize the benefit/risk profile of ADCs, significant efforts have been made to understand their pharmacokinetics (PK) focusing on the absorption, distribution, metabolism, and excretion (ADME) processes, their pharmacodynamics (PD), as well as their dose-effect relationship (PK/PD models) [4].

This use of quantitative mathematical models based on pre-clinical and clinical data to guide decision-making during drug development has become increasingly important in regulatory submissions [7], especially in response to the incentives to make clinical studies more inclusive and to provide optimal doses for more diverse individuals. Physiologically based pharmacokinetic models (PBPK) are particularly well suited to address these needs because of their mechanistic nature, allowing predictions to yet untested scenarios or conditions [8]. These models consider the interplay between physiological parameters and drug properties in relation to ADME processes, and they can predict PK and drug distribution in various tissues. Such models can help translate from in vitro to in vivo studies and from preclinical to clinical studies. They can also provide optimized dosing recommendations in general, specific, or underrepresented populations (age, gender, ethnicities, disease states) and can enhance clinical trial design and success [7].

Several simulation software products, aiming to build and analyze PBPK models, are available. Among them, PK-Sim (Open Systems Pharmacology Suite or OSP) is an open-source software that is increasingly recognized by regulatory authorities. First introduced for small molecules [9], this “whole-body” physiology simulation software was subsequently extended to large molecules [10]. Along with a companion expansion called MoBi, focusing on the cellular and molecular level [11], it has now become possible to implement detailed mechanisms of action at different biological scales. Although PBPK modeling has become a valuable tool in drug development to understand and mechanistically characterize drug exposure in different tissues for many drug modalities [4], very few PBPK models have been published for ADCs to date [12, 13] and none in the OSP suite, to our knowledge. In this study, we aim at developing a PBPK model for an ADC in OSP and at qualifying it with extensive clinical data. Our goal was to capture the PK of an anti-claudin 18.2 ADC (SHR-A1904), which is currently in phase I of clinical development (Hengrui). Claudin 18.2 (CLDN18.2) is a membrane-bound target mostly expressed in nonmalignant stomach epithelium cells and often highly expressed on the tumor cell surface during malignant transformation [14–16].

2 | Methods

2.1 | Dataset

SHR-A1904 is an ADC composed of a humanized CLDN18.2 IgG1 mAb (SHR-1903), a linker, and a payload (SHR169106), with an average DAR of 4. PK of SHR-A1904 was assessed in three phase I studies, including SHR-A1904-I-101 (NCT04877717), SHR-A1904-I-102 (NCT04928625) and SHR-A1904-I-104 (NCT05277168). SHR-A1904 was administered by intravenous (IV) bolus to patients with gastric or pancreatic cancer once every 3 weeks (Q3W).

PK data for the ADC and the deconjugated payload was available in plasma. The full dataset contained nine IV bolus administrations of multiple escalating doses (from 0.6 to 8 mg/kg) in 109 patients with 908 observations for the ADC and 824 observations for the payload (Table 1).

2.2 | Model Development

2.2.1 | Model Input Parameters

The workflow of the analysis is presented in Figure 1. The model input parameters are presented in Table 2. All parameters were assumed to be identical for the ADC and the naked antibody (nAb) except for the molecular weight (MW). In the model, we did not explicitly model all DAR species. Instead, we modeled the molecules with a DAR > 0 as the representative “ADC” molecule with an average DAR equal to 4. We assumed that one ADC released 4 payload molecules and 1 nAb molecule upon deconjugation in plasma or upon endosomal degradation. Hence, the MW between the average ADC and the nAb differed by four times the MW of the payload. During model development, deconjugation was incorporated for the ADC only with a

TABLE 1 | Details of the three clinical phase I studies for SHR-A1904.

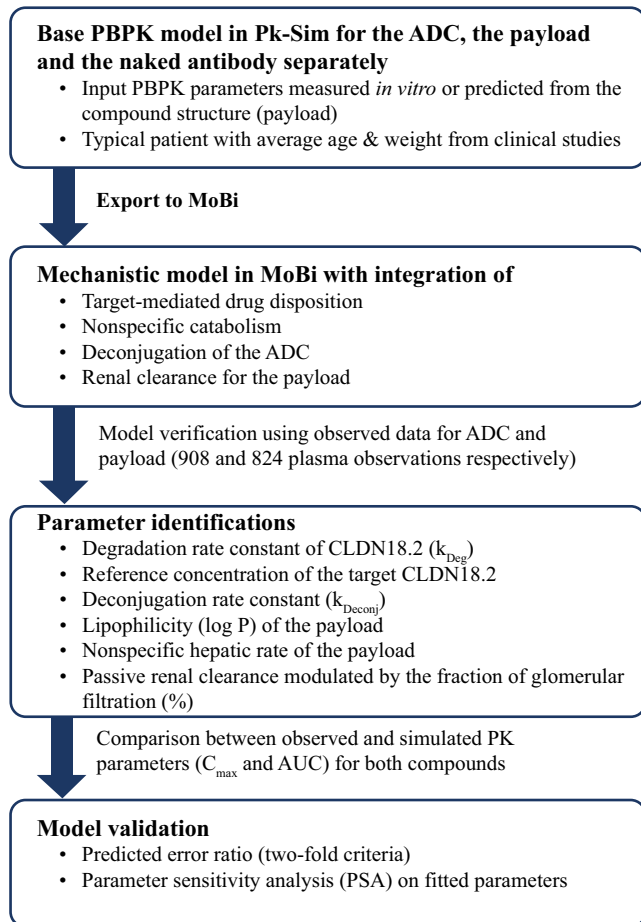
	SHR-A1904-I-101	SHR-A1904-I-102	SHR-A1904-I-104	Total
Cancer type	Gastric cancer	Pancreatic cancer	Gastric, pancreatic and lung cancer	—
Number of patients	72 (mostly Asian)	22 (mostly Asian)	15 (mostly Caucasian)	109
Doses in mg/kg (number of patients per dose group)	0.6 (<i>n</i> = 2), 1.2 (<i>n</i> = 3), 2.4 (<i>n</i> = 3), 3.6 (<i>n</i> = 4), 4.8 (<i>n</i> = 13), 6.0 (<i>n</i> = 16) and 8.0 (<i>n</i> = 31) mg/kg	0.6 (<i>n</i> = 3), 1.2 (<i>n</i> = 3), 2.4 (<i>n</i> = 6), 3.6 (<i>n</i> = 3), 4.8 (<i>n</i> = 3), 6.0 (<i>n</i> = 3) and 8.0 (<i>n</i> = 1) mg/kg	0.6 (<i>n</i> = 3), 0.9 (<i>n</i> = 3), 1.8 (<i>n</i> = 3), 3.6 (<i>n</i> = 3), 4.8 (<i>n</i> = 3) mg/kg	From 0.6 to 8 mg/kg
Age (years)	57 (10)	62 (8)	62 (10)	59 (10)
Weight (kg)	56 (10)	57 (11)	76 (22)	59 (14)
Occurrence of ADA ^a Positive patients (%)	4 (5.6)	4 (18.2)	NA	8 (7.3)
Number of patients with strong CLDN18.2 expression (2+/3+ > 50%) ^b	40	9	10	59
Ethnicity (%)				
Asian	72 (100)	22 (100)	1 (6.7)	95 (87.1)
White	—	—	14 (93.3)	14 (12.8)

Note: Data are mean (SD) or percentage of patients (%).

Abbreviation: NA, not available.

^aAnti-drug antibody.

^bIHC (Immunohistochemistry) results: 2+ and 3+ for staining intensity of the target in 50% of cells.

**FIGURE 1** | Workflow of the analysis.

certain rate described by the parameter k_{Deconj} . Therefore, ADC and nAb were not fully identical as the nAb did not undergo deconjugation, which contributed significantly to the shorter ADC half-life (deconjugation half-life~100h).

CLDN18 is known to occur in two isoforms (CLDN18.1 and CLDN18.2), generated by alternative splicing of exons 1a and 1b, respectively [15]. It is a membrane-bound target, which is mostly expressed in the stomach (CLDN18.2) and lungs (CLDN18.1) in normal tissues [14, 15]. The PK-Sim gene expression library includes large-scale human gene expression data from publicly available sources, such as the ArrayExpress database from the whole genome and the Expressed Sequence Tag (EST) database [17, 18]. These two databases were consistent with the expected expression profile of CLDN18, as the highest concentration in healthy tissues was found in the stomach (used as a reference with 100% relative expression), and then in the lungs (20% relative expression). Even though the ArrayExpress database and the EST database provided similar expression patterns for the organs with a high CLDN18 expression, the ArrayExpress database was selected as it covered a more extensive list of organs with a low expression.

Since CLDN18 is membrane-bound [15] and no expression in plasma was reported in the literature, target localization was selected as interstitial. As the reference concentration in the stomach was unknown, it was first set to a plausible value and subsequently optimized in the average virtual individual.

Based on *in vitro* data, the ADC (and the nAb) affinity for CLDN18.2 and FcRn was set to 2.67 nmol/L and 3.43 μ mol/L, respectively.

TABLE 2 | Input PBPK parameters for the ADC, the nAb and the payload.

Analyte	SHR-A1904	SHR-1903 nAb	SHR169265
Compound type	ADC	Classical mAb (IgG1)	Small molecule
Molecular weight (g/mol)	151.000 ^a	148.870 ^a	533.56 ^b
Target type	Membrane-bound (stomach and lung)		—
Target localization	Interstitial		—
Target reference concentration (nmol/L) in the organ expressing it the most (stomach)	10.00 ^c		—
k_{off} target (1/min)	1.00 (default PK-Sim)		—
k_d target (nmol/L)	2.67 (In vitro ^b)		—
Radius (nm)	5.0 (default PK-Sim)		—
k_d FcRn endosomal space (μmol/L)	3.43 (In vitro ^b)		—
Lipophilicity (log units)	−5 (recommended for large molecules)		2.21 (In silico G+ ^d)
Fraction unbound	1		0.11 (In vitro ^b)
PKA (base)	—		2.93 (In silico G+ ^d)
Solubility (mg/mL)	9999 (recommended for large molecules)		0.05 @ pH 7.02 (In silico G+ ^d)
Blood/Plasma concentration ratio	—		0.67 (In vitro ^b)
Distribution model	Two pores theory (21)		Rodgers and Rowland (23)
Total hepatic clearance	—		Nonspecific hepatic clearance = 0.37 L/h/kg ^b Nonspecific hepatic clearance rate constant = 300 1/min

^aThe 2134.24 Da (g/mol) difference represents DAR × MW_Payload, DAR being equal to 4 here and MW_Payload to 533.56 Da.

^bIn house data from Hengrui.

^cPK-Sim/MoBi library.

^dADMET predictor (3).

For the payload, a few parameters were derived *in silico* with a QSAR approach by the ADMET predictor [19]. The distribution model of Rodgers and Rowland [20] for basic compounds was chosen in PK-Sim for the payload. A nonspecific hepatic clearance was estimated empirically from a previously developed PopPK model for this compound [21]. It was known that the payload also underwent renal and biliary elimination according to preclinical experiments; however, no precise value was measured. Therefore, no renal clearance was initially implemented.

2.3 | PBPK Model in MoBi With TMDD and Deconjugation

2.3.1 | Model Structure

The PK of mAbs is usually described by linear and nonlinear elimination mechanisms with respect to dose. Linear PK is governed by nonspecific pinocytosis/endocytosis and by the FcRn-mediated recycling system. For wild-type antibodies, recycling by the FcRn receptor saturates only at high exposure values, and this threshold is generally not reached within the therapeutic exposure ranges. When nonspecific endocytosis and FcRn recycling are dominant, they result in a dose-proportional

exposure and a constant half-life. In parallel, nonlinear PK of mAbs is governed by the target expression, which modulates the rate of internalization of target-antibody complexes and thus the catabolism and recycling of antibodies. Slower clearance is usually observed at higher doses due to the saturation of the target-mediated clearance, a phenomenon also called Target Mediated Drug Disposition (TMDD) [22, 23]. To correctly describe ADC and payload PK in plasma, these different processes need to be considered in the modeling by implementing different “biochemical reactions” describing the mechanism of action of ADCs. Ideally, as reviewed by Iamele et al. [24], the intravenously injected ADC should remain stable and not cleave or be degraded while in circulation. When the ADC encounters the target at the surface of the tumor cells, it should bind to the antigen and be internalized by these cells via the endosome-lysosome pathway. Then, the payload should be released within the cytoplasm (thanks to linker cleavage or ADC degradation) to enter the cell nucleus and to interfere with DNA to confer cytotoxicity. Here, we assume, as described in Figure 2, that:

1. The target CLDN18.2 has a zero-order synthesis rate and a first-order degradation rate constant (k_{Deg}) across all organs. However, the baseline CLDN18.2 concentration is different for each organ (due to expression profile).

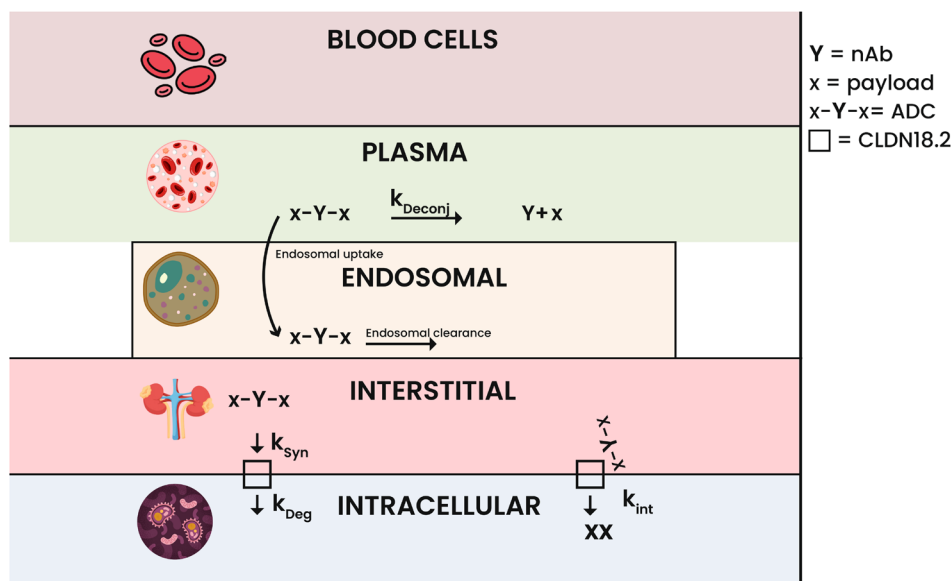


FIGURE 2 | Model structure and corresponding parameters of SHR-A1904 in MoBi.

2.7 | Parameter Sensitivity Analysis of the Model

The model parameter sensitivity analysis (PSA) is the evaluation of whether and to which extent variation in a parameter can influence the drug distribution and elimination following administration. This consists of varying one parameter value while keeping all the other parameters unchanged. All PSA simulations were performed at the dose level of 4.8 mg/kg with the representative individual.

3 | Results

3.1 | PBPK Model of the Three Compounds in PK-Sim

The first PBPK model was built in PK-Sim with SHR-A1904-I-101 and SHR-A1904-I-102 clinical studies at this initial modeling step as described in detail in Material and Methods S1. Therefore, ADC observed data from these two clinical studies were used for model development. Default values of PK-Sim or measured values were used without optimization as reported in Table 2. Overall, the PBPK model was able to capture the ADC concentration-time profile (Figure S1); with this initial parameterization, clearance of the ADC was underestimated, suggesting that additional clearance mechanisms, such as TMDD and/or deconjugation, have a significant impact on the PK and needed to be implemented.

3.2 | Optimizations in MoBi

At this step of model optimization, the observed data from the SHR-A1904-I-104 clinical study in solid cancer patients were appended. The updated dataset included 109 patients in total. The updated dataset confirmed the underestimation of the ADC's clearance within a broader patient population.

As a next step, a series of parameters' optimizations was initiated for several parameter values related to the ADC: k_d FcRn endosomal space, Radius, k_d target, k_{off} target, target reference concentration, k_{int}/k_{Deg} ratio, and deconjugation rate constant K_{Deconj} (description of the parameters and corresponding equations in Material and Methods S2). Finally, three parameters

were identified as sensitive for the ADC's PK: k_{Deg} , the reference concentration of the target CLDN18.2, and k_{Deconj} .

The first optimizations in MoBi showed that some patients were "outliers" and had an increased clearance. This was explained by the presence of anti-drug antibodies (ADAs) in those patients. Indeed, the presence of ADA was identified as a significant covariate on the clearance according to the previous PopPK model [21], with an 80% increase in ADA+ patients. The decision was then taken to remove ADA+ patients to eliminate this important confounding factor and to shrink the unexplained between-subject variability by keeping the ADA- patients only. In addition, ADA- patients were more representative of TMDD, as an identifiability issue might occur between TMDD and ADA in ADA+ patients. Of note, as previously reported for other mAbs [27–30], all the ADA+ patients were in the low dose group and no ADA+ patients were reported in the range of therapeutic doses (i.e., ≥ 4.8 mg/kg). Therefore, eight patients were removed from the observed dataset (4 for study 101 and 4 for study 102). Thus, the final dataset intended to be used for model optimization in MoBi included 101 patients.

After optimizations for the ADC, an optimization of the deconjugation rate constant was necessary when integrating the payload's data. As the distribution and elimination of the payload was not well captured, an adjustment of the lipophilicity improved the overall distribution phase. Renal excretion of the payload was added as an elimination pathway, with the optimization of the fraction of the glomerular filtration rate (fGFR) to correctly capture the elimination of the payload.

3.3 | Final Model With Payload Data in MoBi

Among the parameters explored for parameter optimization using all data simultaneously, five were optimized in the final model and one was fixed from a previous calibration, as described in Table 3.

The final PK model adequately captured the data for both the ADC (in red) and the payload (in blue) as observed in Figure 3. A zoom on the first administration is presented in Figure S2.

TABLE 3 | Optimizations of the final model.

Parameter	Start value	Final value (95% confidence interval)
Degradation rate constant of CLDN18.2 (k_{Deg}) (1/h)	1.00	3.00 E-04 (± 5.91 E-03)
Reference concentration of CLDN18.2 (nmol/L)	10.00	12.64 (± 246.95)
Deconjugation rate constant (k_{Deconj}) (1/h)	1.00 E-03	4.72 E-03 (± 3.50 E-04)
Lipophilicity of the payload (log P)	2.21	1.24 (± 0.62)
Nonspecific hepatic clearance rate constant of the payload (1/min)	300.00	300.00 (fixed)
Passive renal clearance ^a as fraction of glomerular filtration rate (%)	10.00	23.76 (± 6.14)

^aA passive renal clearance was added for the payload in MoBi in addition to the hepatic clearance, fGFR corresponding to the fraction of the GFR (glomerular filtration rate) in the equation $d[\text{Payload}]/dt = fGFR \times GFR \times M$ (M being the molar concentration of payload in the plasma compartment of the kidney).

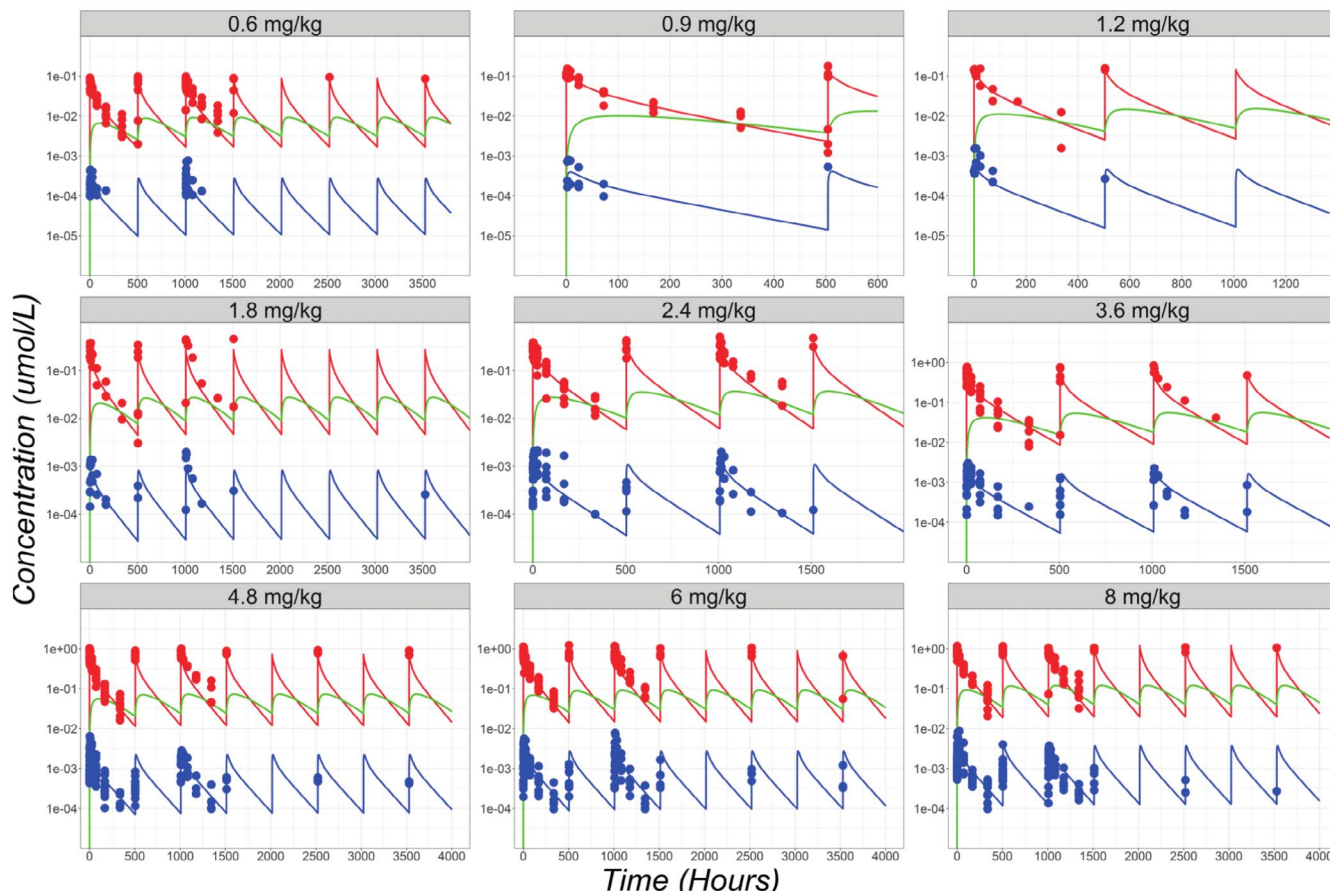


FIGURE 3 | Observed (dots) and simulated (lines) SHR-A1904 (ADC in red), nAb (naked antibody in green), and SHR169106 (payload in blue) concentration time-course in cancer patients following IV administration (Final MoBi model) from studies 101, 102, and 104.

3.4 | Population Simulations

Population simulations were performed for the ADC and the payload using body weight- and age-matching populations in each dose group in clinical trials (Table S1). Physiological parameters (e.g., blood flow) which are known to depend on ethnicity, age, sex, and body weight were generated with their respective distributions [26]. In addition, the between-subject variability of parameters related to the target, TMDD, and deconjugation processes was also included. All these parameters and corresponding mean and SD for population simulations are presented in Table S2. Then, the geometric SD in the predictions were compared to the observed geometric SD for $C_{\max}$ and AUC_{0-504h} of the ADC and the payload (Table S3). The predicted and observed geometric SD were close, suggesting the good predictive performance of the PBPK model, together with the associated between-subject variability, in an Asian population aged between 27 and 84 years old (Figure S4 for the dose of 6 mg/kg).

3.5 | Model Evaluation

Evaluation of the PBPK Model performance was performed as described in Material and Methods S3. Prediction error (PE) ratio for the final model was between 1.07 and 1.50 for $C_{\max}$ of the ADC and between 0.56 and 1.18 for $C_{\max}$ of the payload, with 7 out of 9 predicted values within 0.80–1.25 for the ADC and 6

out of 9 predicted values for the payload. Similarly, all AUC_{0-504h} (first administration) were included in the twofold range, with PE ratio between 0.73 and 1.23 for the ADC and between 0.77 and 1.37 for the payload. More precisely, 8 out of 9 predicted values for the ADC and 5 out of 7 predicted values for the payload were within 0.80–1.25. These PE ratios are illustrated in Figure 4 and Table S3.

3.6 | Parameter Sensitivity Analysis

A PSA was performed on optimized parameters at the end of the analysis. As an example, a sensitivity of -1.0 implies that a 10% increase in the parameters leads to a 10% decrease in the PK parameter value, and a sensitivity of $+0.5$ implies that a 10% increase in the parameters leads to a 5% increase in the PK parameter value. The results (Table S4) showed that the largest impact on $C_{\max}$ of the payload and AUC of the ADC was related to the deconjugation rate constant. In this analysis, a 10% increase in the deconjugation rate constant leads to a 3.5% increase in the $C_{\max}$ of the payload and a 3.2% decrease in the AUC of the ADC. No impact on $T_{\max}$ of any of the six assessed parameters was observed for the ADC and the payload.

A PSA on k_d FcRn was also performed, showing that a 10% increase in k_d FcRn led to a 5.9% decrease in the AUC of the ADC and a 4.3% increase in the $C_{\max}$ of the payload.

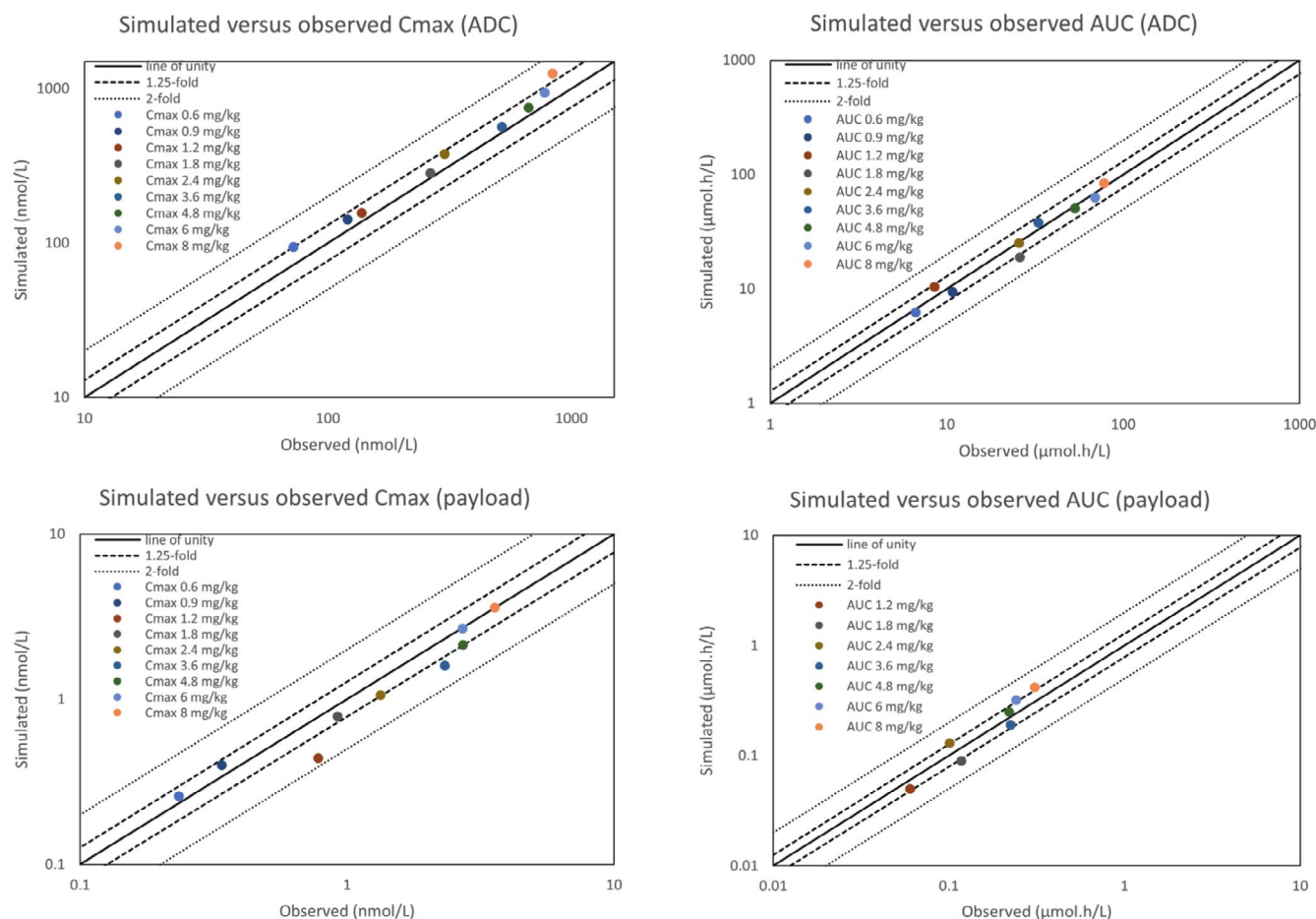


FIGURE 4 | Prediction performance of the model presented as the correlation between simulated versus observed $C_{\max}$ and AUC_{0-t} on the ADC (upper panel) and the payload (bottom panel).

4 | Discussion

In this study, we developed a PBPK model for the ADC SHR-A1904 and the released payload. The model incorporated ADC administration and disposition, binding to the target in healthy tissues, and intracellular processing of the ADC and payload within the endosomal/lysosomal compartment leading to ADC degradation, payload release, and target recycling. These processes were set in MoBi to mimic a detailed mechanism of action. Several parameters were optimized to fit the observed data, firstly for the ADC, subsequently for the payload, and then with all data simultaneously. Population simulations enabled the comparison of the observed between-subject variability at each dose in 100 virtual individuals. Implementing a variability term on the TMDD and deconjugation parameters enabled the model to reasonably predict the PK between-subject variability of both the ADC and payload. These results illustrate further the ability of PBPK modeling to describe ADC and payload PK by considering in vitro parameters and physiology. The deconjugation rate constant appeared to be the most influential parameter, whereas TMDD parameters seemed to have a relatively limited impact on ADC and payload PK.

Although this PBPK model successfully predicted ADC and payload's PK, a few aspects still need to be confirmed in the

future. As mentioned above, six parameters were fitted without validation of these adjusted values. Several parameters' combinations can lead to the same predicted values, with no certainty as to which pair of values is more plausible physiologically. Therefore, although the PK model was deemed adequate, the target reference concentration and value of the deconjugation rate constant need to be confirmed experimentally in the future. Furthermore, while the ADC deconjugation into nAb and payload was incorporated with an average DAR equal to 4, no progressive deconjugation was considered. There was not enough observed data to allow an estimation of each DAR-specific clearance, so the ADC model was considered as the average ADC with all different DAR values. It was assumed that lumping the consecutive deconjugation steps into a single step had negligible to no impact on the PK. Deconjugation, TMDD, and pinocytosis were considered in this model, with deconjugation only impacting total clearance of the ADC but not the clearance of the nAb. Moreover, a few input parameters for the payload were taken from a previously developed PopPK model, highlighting the utility of such top-down modeling prior to PBPK modeling. Nevertheless, these values were further optimized at the end of the present analysis, limiting the need to have a preliminary top-down model. Finally, a few missing experimental in vitro measurements should have ideally been used for payload elimination pathways if available. When adding in vitro interaction parameters of the payload toward cytochromes (CYPs) involved

in its elimination, the current PBPK model can be used to assess its drug–drug interaction.

In this model, only plasma concentrations were analyzed, without predictions of tissue concentrations in tumors. In fact, this work paves the way to a future PBPK-PD model incorporating a tumor compartment, by showing how much physiology only can account for PK of ADC and payload. In addition, two publications [25, 31] suggest the limited contribution of tumors and small volume organs like the heart to total clearance and nonlinearity of mAbs. This confirms the limited need for incorporation of tumor TMDD for correct prediction of plasma concentrations, even though it remains essential for capturing PBPK-PD and exposure-response aspects. Thus, to support the dose optimization with respect to efficacy, the current PBPK model could be leveraged by integrating a tumor compartment to simulate the exposure-response relationship. Unlike more empirical tumor growth inhibition (TGI) modeling reported using the population approach [32–34], a more physiological model structure would increase the mechanistic understanding of the underlying processes related to between-patient variability in target expression, tumor growth, and drug uptake inside tumor cells and the correlations between those variability terms. A better understanding of drug-induced tumor shrinkage could potentially allow the investigation of the conditions (such as baseline target expression and turnover) and treatment combinations under which tumors are more susceptible to developing resistance. Capturing those processes would require the implementation of the impact of baseline target expression and its heterogeneity across and within patients' tumors. By integrating individual patient characteristics and dynamic biomarkers (such as tumor type, tumor size, target expression and ultimately circulating tumor DNA) into therapeutic decision-making, we could move beyond traditional paradigms toward a more tailored and effective approach. This precision-dosing mindset would not only optimize treatment outcomes, but it would also underscore the commitment to individualized care, paving the way for a future where therapies are as unique as the patients they serve.

5 | Conclusion

Our model allowed an accurate prediction of the observed data in cancer patients for an ADC targeting CLDN18.2. Missing or uncertain model parameters were adjusted for the different constants of the TMDD and deconjugation elimination pathways. More case studies are needed for other ADCs to confirm the applicability of PBPK modeling. Nevertheless, the accurate ADC and payload simulations' outcomes for this ADC represent an important first step in the PBPK modeling of ADCs.

Author Contributions

C.Z., S.U., and N.D. wrote the manuscript. N.D., X.D., and A.P. designed the research. C.Z. performed the research. S.W., Y.Z., and N.D. analyzed the clinical data. W.E.A.W. contributed new analytical tools.

Conflicts of Interest

The authors declare no conflicts of interest.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.